

Structured Query Language (SQL)

SQL stands for Structured Query Language.

It is the standard language used to communicate with relational databases. SQL is a declarative language, not a general-purpose programming language. You tell the database:

- **WHAT** you want.
- Not **HOW** to get it.

What Can SQL Do?

SQL can:

- Create databases
- Delete databases
- Create tables
- Delete tables
- Insert records
- Update records
- Delete records
- Search records
- Join tables
- Create indexes
- Create views
- Create stored procedures
- Manage users
- Control permissions
- Manage transactions

Basic SQL Operations

Create database

```
CREATE DATABASE College;
```

Create table

```
CREATE TABLE Student(  
    StudentID INT,  
    Name VARCHAR(100)  
);
```

Insert

```
INSERT INTO Student  
VALUES (1, 'Aadya');
```

Read

```
SELECT *  
FROM Student;
```

Update

```
UPDATE Student  
SET Name='Arpit'  
WHERE StudentID=1;
```

Delete

```
DELETE FROM Student  
WHERE StudentID=1;
```

SQL vs DBMS

SQL	DBMS
A language	Software
Used to communicate with a database	Manages the database

Cannot store data by itself	Stores and manages data
Standardized	Different implementations exist
Examples: SELECT, INSERT	Examples: MySQL, PostgreSQL, Oracle

What is a SQL Dialect?

A dialect is a vendor-specific variation of standard SQL. The core language is shared, but each DBMS adds its own features and syntax.

Major SQL Dialects

Dialect	Characteristics
MySQL	• Open source • Owned by Oracle • Very popular for web applications • Fast and beginner-friendly • Used in many startups and educational environments
PostgreSQL	• Open source • Known for standards compliance • Rich feature set • Excellent for complex queries and advanced data types • Popular in enterprise systems and data-intensive applications
SQL Server	• Developed by Microsoft • Uses the T-SQL dialect • Common in enterprises using the Microsoft ecosystem

Oracle Database	• Developed by Oracle • Uses PL/SQL extensions • Very common in banking, telecom, and large enterprises
SQLite	• Lightweight • Serverless • Entire database is stored in a single file • Common in mobile apps, embedded devices, and local applications

Example of Dialect Differences

Get the first five rows.

MySQL / PostgreSQL

```
SELECT *
FROM Student
LIMIT 5;
```

SQL Server

```
SELECT TOP 5 *
FROM Student;
```

SQL Command Categories

Although SQL is one language, its commands are commonly grouped by purpose.

1. DDL — Data Definition Language

Defines or changes the database structure.

Common commands:

- CREATE
- ALTER
- DROP

- TRUNCATE
- RENAME

2. DML — Data Manipulation Language

Works with the data stored in tables.

Common commands:

- INSERT
- UPDATE
- DELETE

3. DQL — Data Query Language

Retrieves data.

Main command:

- SELECT

4. DCL — Data Control Language

Controls permissions and access.

Common commands:

- GRANT
- REVOKE

5. TCL — Transaction Control Language

Manages transactions.

Common commands:

- COMMIT
- ROLLBACK
- SAVEPOINT

Summary Table

Category	Purpose	Common Commands
DDL	Define database objects	CREATE, ALTER, DROP, TRUNCATE, RENAME

DML	Modify data	INSERT, UPDATE, DELETE
DQL	Retrieve data	SELECT
DCL	Control permissions	GRANT, REVOKE
TCL	Manage transactions	COMMIT, ROLLBACK, SAVEPOINT